

New Search for Dark Matter Free Dwarf Galaxies using Legacy Imaging

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Abstract: Studies of ultra-diffuse galaxies (UDGs) NGC 1052 DF2 and DF4, in addition to the more recent study of FCC 224, confirm little to no dark matter content in these galaxies, an unprecedented discovery which remains controversial due to the challenges it poses to the currently understood galaxy formation models. These well-studied dark matter free UDGs share a pattern of hosting overly-luminous, monochromatic globular clusters; believed to be closely connected to the formation of these galaxies. By parameterizing these globular clusters, a method was developed to guide a search for new dark matter free candidates. The search is still ongoing, though the search method has been developed and one interesting candidate has been found already.

Background

The only known dark matter free dwarf galaxies are NGC-1052 DF2, DF4 (van Dokkum et al. 2018, 2019) and FCC 224 (Buzzo et al. 2025). These dwarf galaxies exhibit a pattern of hosting overly luminous, monochromatic globular clusters (GCs; Shen et al. 2021, Tang et al. 2025). Publicly available Legacy Imaging Survey images these objects with great clarity, though the ground based photometry for GCs pose some challenges such as shredding, crowding, and galaxy light contamination. A large part of this project was focused on these challenges and characterizing the interesting GCs found within the three known dark matter free dwarf galaxies by magnitude, color, and concentration to finalize our search parameters. In our method, interesting dwarf galaxies are first selected by eye if they share properties with the known dark matter free dwarfs, then the candidate GCs within the system are confirmed quantitatively with our finalized GC parameters. If numerous candidate GCs are found within the system, galaxy modeling and light subtraction is performed to do residual photometry, the results of which give clues to whether or not the galaxy may be a strong dark matter free candidate.

Methods

Legacy’s DR10 photometry data was collected for DF2, DF4, and FCC 224 in order to determine the selection parameters, including magnitude vs color, magnitude vs concentration, and color vs concentration. This allowed us to define the parameters within which these clusters are likely to exist: 0.75 arcsecond aperture magnitude in the g-band between 22.5 - 24, concentration (0.5-1 arcsec aperture difference) between 0.7 - 1.3, and g-i color between 0.75-1.2. These finalized selection parameters allow us to quantitatively confirm the possible GC candidates within an interesting dwarf system after initial selection by eye (overly luminous, monochromatic GCs) by using SQL search. If a UDG is visually interesting and has numerous candidate GCs according to the search, galaxy light subtraction is done using AstroPhot (Stone et al. 2023) and photometry on candidate GCs is performed on the residual image. The obtained values are compared to known literature values of the GCs within the known dark matter free dwarf galaxies to guage the possibility of the galaxy containing no dark matter.

Results and Future Research

The search for a strong dark matter free candidate is still ongoing, though the search method shows initial promise with the preliminary study of dwarf galaxy VCC 2084 (Figure 4). Additional spectroscopy must be done on this galaxy in order to determine redshifts and ultimately dark matter content. In the meantime the search for additional candidates continues using the developed method.

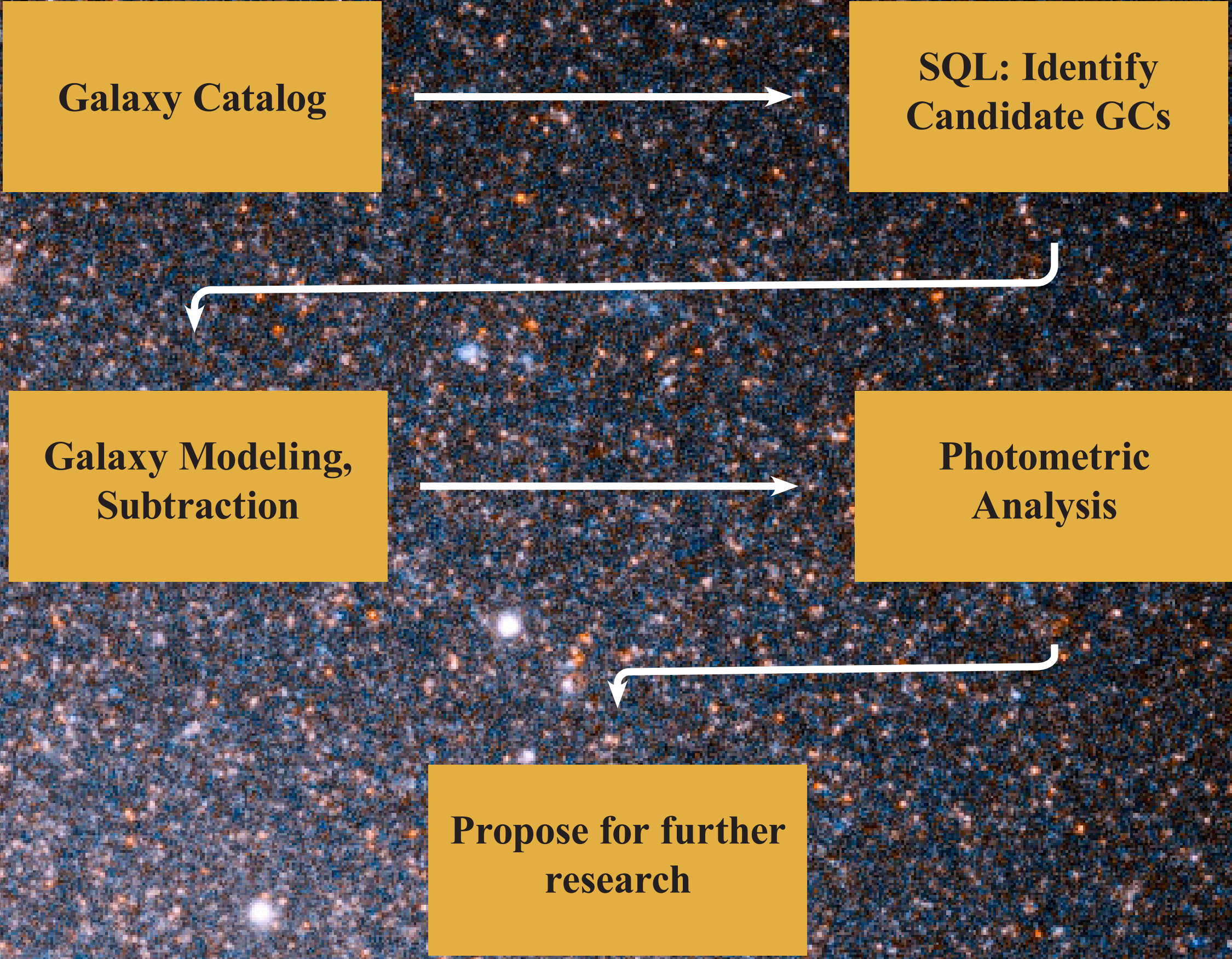


Figure 1: Illustrating the method developed to search for new dark matter free dwarf galaxy candidates.

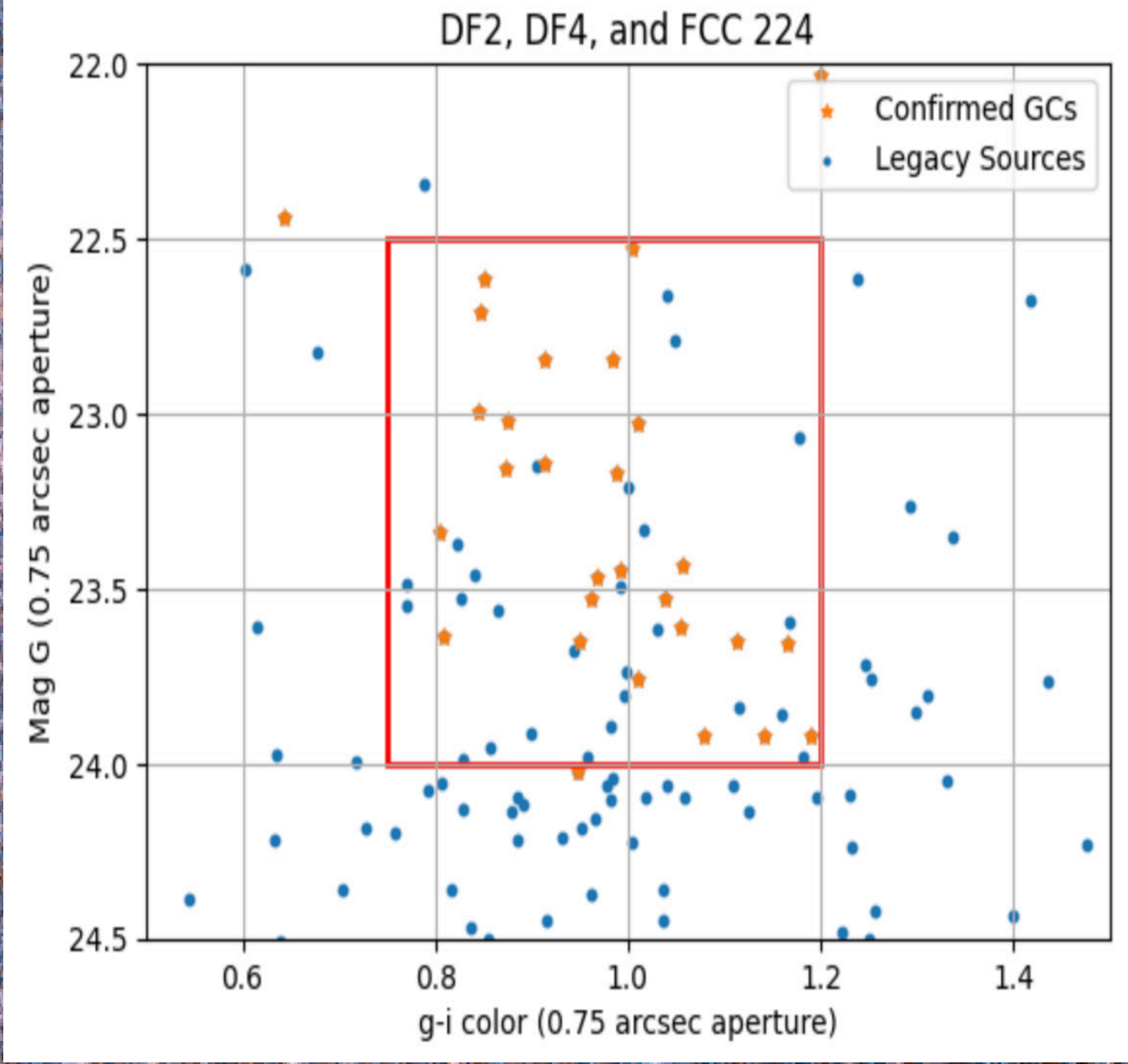


Figure 2: Plot showing magnitude vs color, both using 0.75 arcsecond apertures. Confirmed GCs are plotted alongside all sources within the search radius. Selection box represents finalized parameters which define an interesting GC. Other plots included magnitude vs concentration and concentration vs color.

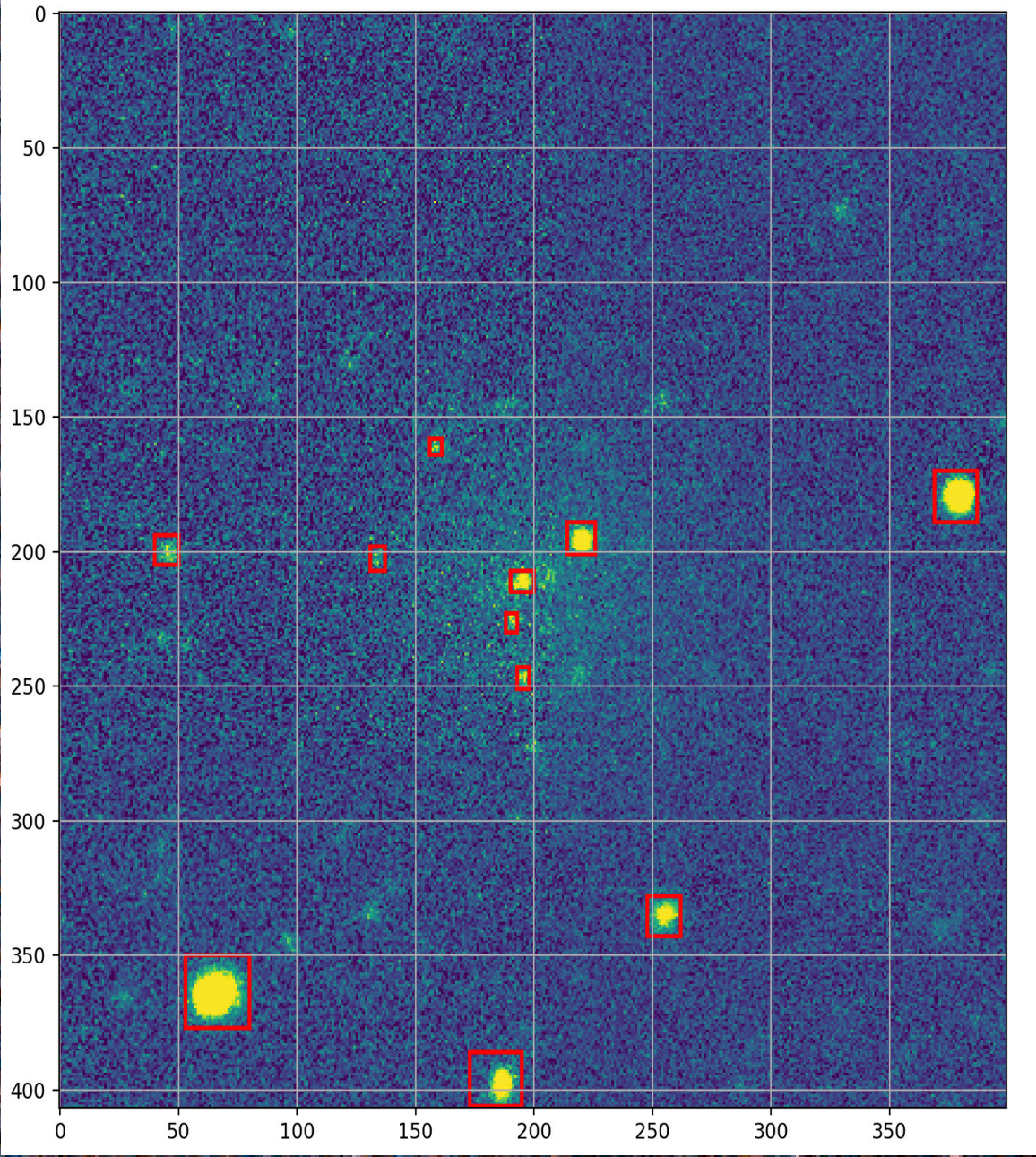


Figure 3: An example of the galaxy subtraction process, necessary for improved photometry of candidate GCs. Galaxy model is used to determine galaxy luminosity. Subtracted magnitudes are used to determine luminosity of GC system.

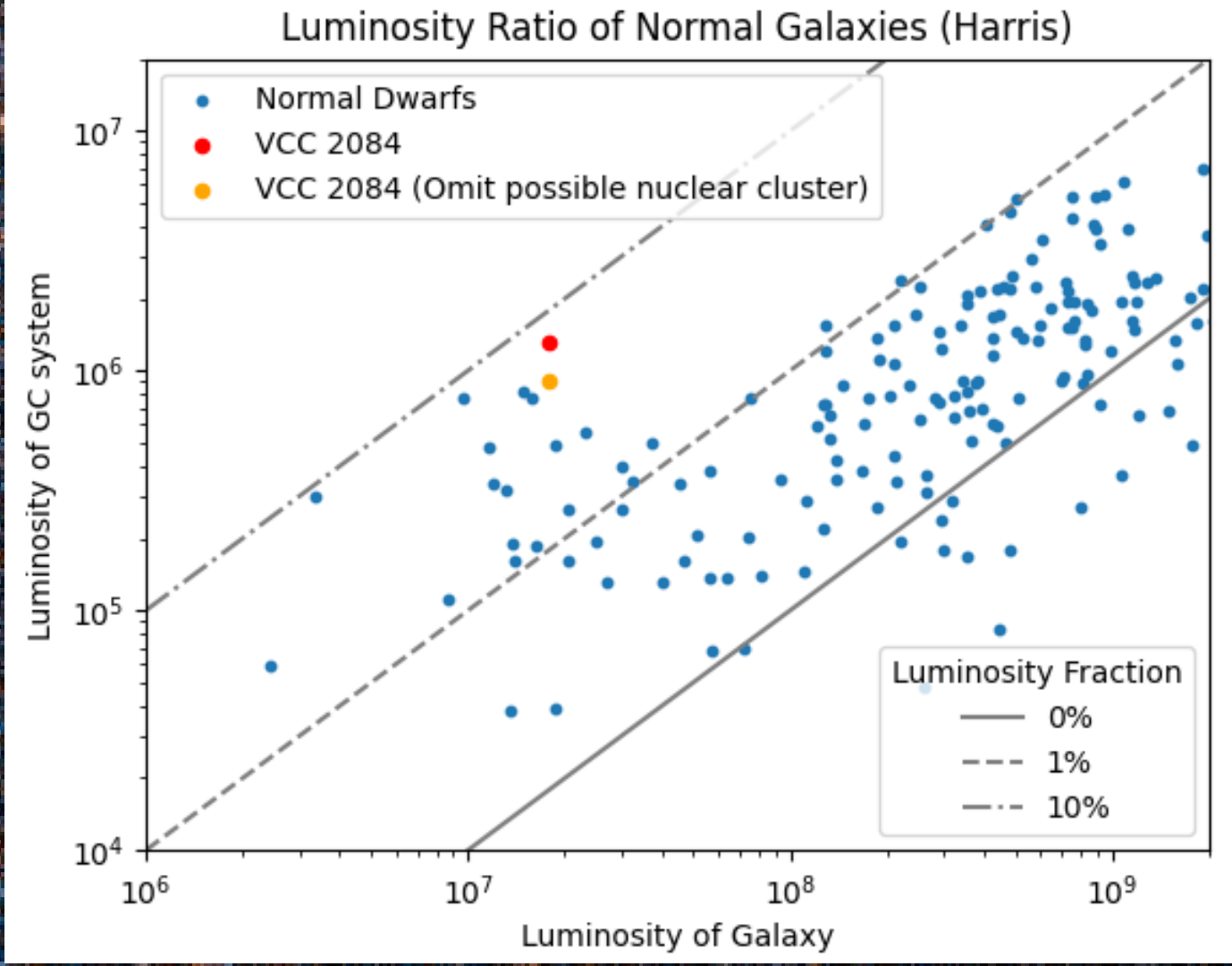


Figure 4: Pictured is VCC 2084, picked visually due to its low luminosity and monochromatic, luminous globular clusters. Determining the luminosity ratio of VCC 2084 and comparing it with other dwarf galaxies of similar type shows that VCC 2084 is a potential outlier, constituting further research in order to determine dark matter content.

References

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